

MONIL ARORA

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Education

University of British Columbia, Vancouver

Expected 2027

Bachelor of Applied Science in Electrical Engineering

Relevant Coursework: Machine Learning with Engineering Applications, Digital Systems Design, Computing Systems I, Signals and Systems, Introduction to VLSI Systems

Work Experience

Laser Zentrum Hannover (LZH)

Jan – Jul 2026

Research Intern

Hannover, Germany

- Developed **STM32** flight firmware in **C++/FreeRTOS** across two targets (**F401CC** dev board, **F103C8** mission PCB), packetizing telemetry into **CCSDS SPP** frames **SLIP**-encoded over **RS-422** through a **Lantronix UDS2100** bridge, with a ported **COP-1 FARM-1** receiver publishing **CLCW** in the TM frame **OCF**, a **LittleFS** boot counter and RTC time persistence in flash, and a priority **bypass (BD) command queue**; diagnosed and fixed an **IWDG** reset mid image-transfer by relocating task check-ins inside the transmission loop.
- Extended **YAMCS** Mission Control with custom **Java data links** for **SLIP**-over-**UDP** telemetry and **CCSDS TC** uplink, **COP-1 FOP**-compliant delivery, an **XTCE mission database** decoding **APID 130 TM / 100 TC** with threshold alarms and telemetry-based **command verifiers**, a custom command releaser routing an uplink **firmware-update** command off the spacecraft path, and a **yamcs-client** script automating **COP-1** reboot recovery.
- Implemented end-to-end **image downlink** with an 8 KB SRAM buffer sending 79-byte **APID 131 / VCID 1** chunks on firmware, and a Python **WebSocket** reassembler reconstructing JPEGs into **YAMCS** object storage; validated against a **PyQt6** payload simulator, **Python TCP-UDP** bridge, and a **hardware-in-the-loop Basler GigE** camera.

Sharang Shakti Pvt Ltd

May – Jun 2024

Embedded Signal Processing Engineer

Gurugram, India

- Contributed to the **digitalization of an analog radar system** by bringing up an **AD9910 DDS** evaluation board and generating high-frequency transmit waveforms via profile registers, **RAM playback**, and the **digital ramp generator**; wrote Python to build **14-bit DAC sample tables** and stabilized the reference-clock **PLL** to improve resolution and reduce **phase noise**.

Projects

45 nm Dot-Product Accelerator | *SystemVerilog, Cadence Genus, Innovus, ModelSim*

Sep – Dec 2025

- Designed a four-element **MAC dot-product accelerator** in **SystemVerilog** and took it **RTL-to-GDSII** on a **45 nm** PDK with **Genus** synthesis, **Innovus** place-and-route, and **clock tree synthesis** (**104 ps** skew, 1,320 flops).
- Characterized power from real switching activity, running **gate-level simulation** with back-annotated **SDF** delays in **ModelSim** and converting **VCD to SAIF**, which corrected a 1.36 mW estimate to **68.7 μ W**.

Self-Balancing Robot | *C++, Arduino, ESP32, IMU, PID, Embedded Systems*

Jan – Apr 2025

- Designed and built a two-wheeled **self-balancing robot** using an **Arduino Nano 33 BLE**, a **BMI270 IMU**, and **motor drivers**, implementing a real-time **PID control loop** with **complementary-filter** sensor fusion and **PWM** motor drive for tilt stabilization up to **45°**; validated through **oscilloscope analysis**, achieving approximately **30% improvement in stability**.
- Developed an **ESP32 BLE** remote transmitting drive commands and displaying live RPM, tilt angle, and sonar distance on an **LCD**.

Deep Neural Network Accelerator | *Verilog, C, Nios II, FPGA*

Oct – Nov 2024

- Implemented a **deep neural network accelerator** on an embedded **Nios II system** using the **DE1-SoC** board for MNIST classification with **Q16.16 fixed-point arithmetic** and **ReLU** activation.
- Developed a **system-on-chip (SoC)** with custom **Avalon** dot-product accelerators and an **SDRAM interface**, applying **Quartus timing constraints** to ensure stable operation at **50 MHz**.

Reflow Oven Controller | *Assembly, 8051*

Jan – Feb 2024

- Developed a **microcontroller-based reflow oven controller** using the **Nuvoton N76E003 (8051 core)** in **assembly language**, automating a multi-stage **soldering temperature profile** (a six-state machine) from **25°C** to **240°C** to within $\pm 3^\circ\text{C}$, with **automatic safety shutdowns and alerts** and a **real-time PC monitoring interface**.

Technical Skills

Languages: C/C++, Python, Java, Assembly, Verilog/SystemVerilog

Tools: YAMCS, FreeRTOS, PlatformIO, Maven, Quartus, ModelSim, CrossIDE, Cadence, MATLAB, LTspice, Git

Hardware: STM32, ESP32, EFM8, Arduino, DE1-SoC, Raspberry Pi, IMU Sensors, Motor Drivers

ML/Data: PyTorch, scikit-learn, NumPy, Matplotlib, CVXPY